

SIMATS ENGINEERING
CO3 - ASSESSMENT TOOL - 2: Case Study

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Dave's Taxonomy: Precision (Level 3)
Question Count: 3

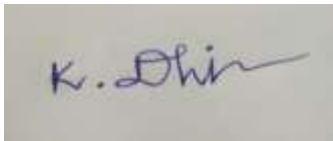
Total Marks: 30

Code of Conduct

I **K.Dhilip/192365013** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Assessment Tasks - Case Study

1. Case Study 1: Smart Retail Inventory Management System

A retail chain implements a cloud-based inventory system connected to store devices. Clients (POS systems and dashboards) communicate via APIs to centralized cloud services. Cloud storage stores product data, transaction logs, and analytics datasets. The system uses platform services for analytics and demand forecasting. Secure access is maintained through VPNs, firewalls, and API gateways. This solution integrates infrastructure, services, and web applications for efficient operations.

2. Case Study 2: Cloud-Based Document Collaboration System

An enterprise deploys a document collaboration tool similar to Google Docs. Users access the platform via browsers, enabling real-time editing and sharing. Documents are stored in distributed cloud storage with version control mechanisms. Web APIs enable synchronization across multiple clients and devices. Network infrastructure ensures low latency and high availability globally. Security includes encryption, identity management, and access permissions for collaboration.

3. Case Study 3: Online Food Ordering Platform

A startup builds a cloud-based food ordering system accessible via web browsers and mobile clients. Frontend clients interact with backend services through RESTful Web APIs hosted on a cloud platform.

Cloud storage is used to store menu images, invoices, and customer data securely. The infrastructure uses virtual machines and managed Kubernetes for scalability and reliability. Security is enforced through authentication tokens, HTTPS, and role-based access control. The system demonstrates integration of clients, APIs, storage, and platform services in real time.

Case Study Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Case	25%	Clearly identifies all technical and business requirements	Identifies most requirements	Basic understanding only	Poor understanding of case
Application of Concepts	25%	Applies suitable cloud concepts and tools effectively	Good application with minor issues	Partial application	Weak or incorrect application
Analysis	20%	Strong analysis with clear trade-off reasoning	Reasonable analysis	Limited analysis	Minimal analysis
Solution Design	20%	Solution is feasible, practical, and well justified	Reasonable solution with minor gaps	Basic solution	Unclear solution
Clarity	10%	Well-structured and easy to follow	Mostly clear	Somewhat unclear	Poorly organized

1. Case Study 1: Smart Retail Inventory Management System

A retail chain implements a cloud-based inventory system connected to store devices. Clients (POS systems and dashboards) communicate via APIs to centralized cloud services. Cloud storage stores product data, transaction logs, and analytics datasets. The system uses platform services for analytics and demand forecasting. Secure access is maintained through VPNs, firewalls, and API gateways. This solution integrates infrastructure, services, and web applications for efficient operations.

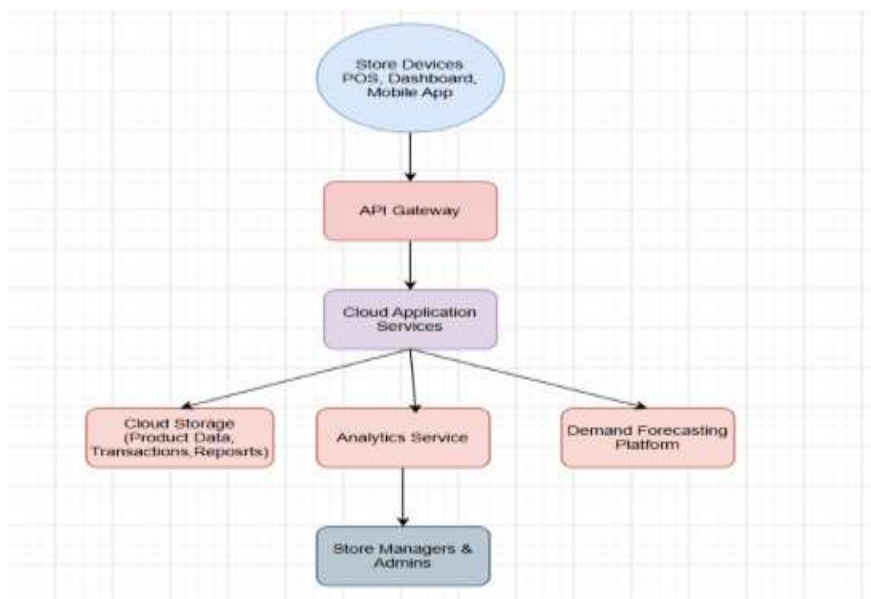
A retail chain wants to manage its inventory efficiently across multiple stores. To achieve this, it implements a cloud-based inventory management system that connects all store devices to centralized cloud services. Point of Sale (POS) systems, employee dashboards, and mobile applications communicate with the cloud using secure APIs. This enables real-time inventory tracking, faster decision-making, and improved customer service.

System Architecture

The cloud-based inventory system consists of the following components:

- **Clients:** POS systems, web dashboards, and mobile applications used by store employees and managers.
- **API Gateway:** Securely receives requests from clients and forwards them to cloud services.
- **Cloud Application Services:** Process inventory updates, sales transactions, stock management, and order requests.
- **Cloud Storage:** Stores product details, transaction logs, customer purchase history, and analytics data.
- **Platform Services:** Analyze sales data, generate reports, and forecast future product demand.
- **Security Components:** VPN, firewall, authentication, and API gateway protect the system from unauthorized access.

Architecture Flow



Role of Cloud Components

Clients

Clients include POS systems, dashboards, and mobile applications. They allow employees to:

- Record sales transactions.
- Check product availability.
- Update stock information.
- View reports and inventory status.

APIs

APIs act as a communication bridge between store devices and cloud services.

They help to:

- Send sales and inventory data securely.
- Retrieve product information.
- Update stock levels in real time.
- Connect multiple stores to the same cloud platform.

Cloud Storage

Cloud storage securely stores:

- Product information
- Inventory records
- Sales transactions
- Customer purchase history
- Analytics datasets

It provides high availability, automatic backup, and easy scalability.

Platform Services

Cloud platform services perform advanced operations such as:

- Sales analytics
- Demand forecasting
- Inventory optimization
- Report generation
- Business intelligence dashboards

These services help managers make better business decisions.

Security Features

The system protects business data using several security mechanisms:

- **VPN (Virtual Private Network):** Creates a secure connection between stores and the cloud.
- **Firewall:** Blocks unauthorized network access.
- **API Gateway:** Authenticates and authorizes API requests before forwarding them.
- **HTTPS Encryption:** Protects data while it is transmitted over the internet.
- **Role-Based Access Control (RBAC):** Ensures employees can access only the information required for their roles.
- **Multi-Factor Authentication (MFA):** Adds an extra layer of security for administrators.

These measures help protect sensitive inventory and sales data.

Benefits of the Solution

- Real-time inventory updates across all stores.
- Faster communication between store devices and cloud services.
- Improved demand forecasting using cloud analytics.
- Reduced stock shortages and overstock situations.
- Secure storage of business data with automatic backups.
- Easy scalability as the business expands to new locations.
- Better decision-making through real-time reports and dashboards.
- Lower infrastructure maintenance costs compared to traditional systems

2. Case Study 2: Cloud-Based Document Collaboration System

An enterprise deploys a document collaboration tool similar to Google Docs. Users access the platform via browsers, enabling real-time editing and sharing. Documents are stored in distributed cloud storage with version control mechanisms. Web APIs enable synchronization across multiple clients and devices. Network infrastructure ensures low latency and high availability globally. Security includes encryption, identity management, and access permissions for collaboration.

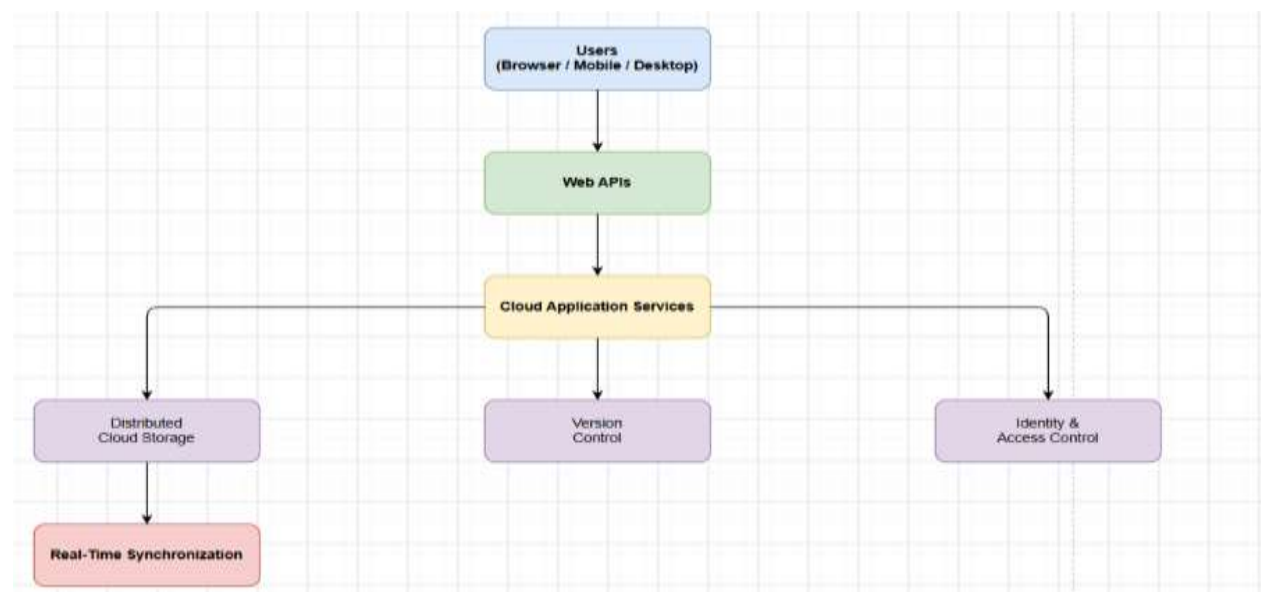
An enterprise wants to provide employees with a cloud-based document collaboration system similar to Google Docs. The system allows multiple users to create, edit, and share documents in real time from anywhere using a web browser or mobile device. All documents are securely stored in the cloud, ensuring easy access, automatic saving, and seamless collaboration.

System Architecture

The cloud-based document collaboration system consists of the following components:

- **Clients:** Web browsers, mobile apps, and desktop applications used by employees.
- **Web APIs:** Synchronize document changes between multiple users and devices in real time.
- **Application Services:** Manage document editing, sharing, user authentication, and collaboration features.
- **Distributed Cloud Storage:** Stores documents securely with automatic replication and version control.
- **Network Infrastructure:** Uses Content Delivery Networks (CDNs), load balancers, and multiple cloud regions to provide fast and reliable access worldwide.
- **Security Services:** Protect user data using encryption, identity management, and access permissions.

Architecture Flow



Role of Cloud Components

Clients

Users access the platform through:

- Web browsers
- Mobile applications
- Desktop applications

They can:

- Create documents
- Edit documents in real time
- Share files with others
- View changes instantly

Web APIs

Web APIs enable communication between clients and cloud services.

They help to:

- Synchronize document updates across multiple devices.
- Save changes automatically.
- Support real-time collaboration.
- Keep all users viewing the latest version of the document.

Distributed Cloud Storage

Cloud storage securely stores:

- Documents
- Images
- Shared files
- User settings
- Version history

Since data is distributed across multiple servers, documents remain available even if one server fails.

Version Control

Version control keeps track of every change made to a document.

It allows users to:

- View previous versions.
- Restore accidentally deleted or modified content.
- Track who made each change.
- Collaborate without losing important information.

Network Infrastructure

The system ensures fast and reliable performance through:

- **Content Delivery Networks (CDNs):** Deliver files from servers closest to users, reducing latency.
- **Load Balancers:** Distribute user requests across multiple servers.
- **Multiple Cloud Regions:** Improve availability and provide disaster recovery.
- **Auto-Scaling:** Automatically adds resources when many users access the platform simultaneously.

This infrastructure provides a smooth experience for users around the world.

Security Features

The collaboration platform protects documents using several security mechanisms:

- **Encryption:** Protects documents during storage and transmission using strong encryption standards.
- **Identity Management (IAM):** Verifies user identities before granting access.
- **Role-Based Access Control (RBAC):** Allows administrators to assign permissions such as View, Edit, or Comment.
- **Multi-Factor Authentication (MFA):** Adds an extra layer of security during login.
- **Audit Logs:** Record document activities for monitoring and compliance.

These features ensure that only authorized users can access and modify shared documents.

Benefits of the Solution

- Real-time collaboration among multiple users.
- Automatic document synchronization across all devices.
- Secure cloud storage with backup and version control.
- Fast access from anywhere using browsers or mobile apps.
- High availability through distributed cloud infrastructure.
- Improved productivity with instant sharing and editing.
- Easy scalability as the number of users grows.
- Reduced IT maintenance compared to traditional document management systems.

3. Case Study 3: Online Food Ordering Platform

A startup builds a cloud-based food ordering system accessible via web browsers and mobile clients. Frontend clients interact with backend services through RESTful Web APIs hosted on a cloud platform. Cloud storage is used to store menu images, invoices, and customer data securely. The infrastructure uses virtual machines and managed Kubernetes for scalability and reliability. Security is enforced through authentication tokens, HTTPS, and role-based access control. The system demonstrates integration of clients, APIs, storage, and platform services in real time.

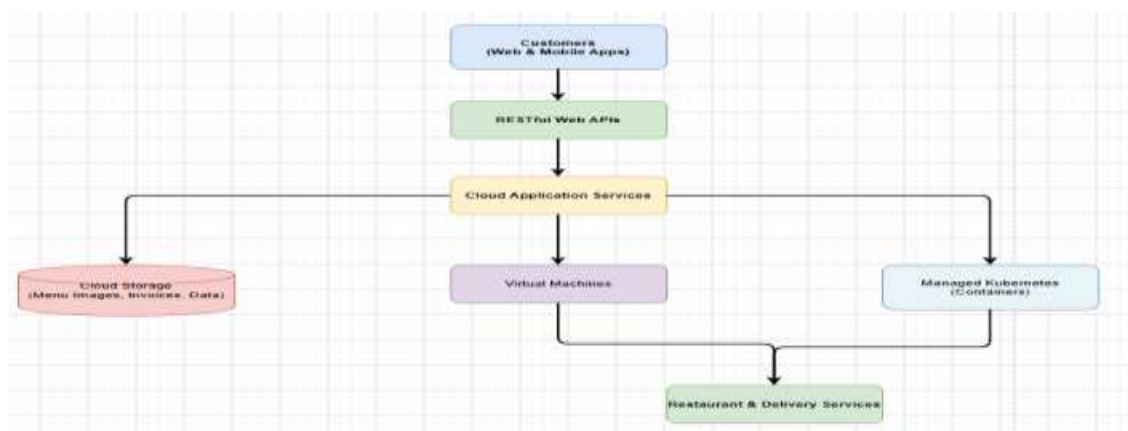
A startup develops a cloud-based online food ordering platform that allows customers to order food through web browsers **and** mobile applications. The frontend communicates with backend services using RESTful Web APIs hosted on a cloud platform. Cloud storage securely stores menu images, invoices, and customer information. The system uses Virtual Machines (VMs) and Managed Kubernetes to ensure scalability, reliability, and smooth performance. Strong security measures such as authentication tokens, HTTPS, and Role-Based Access Control (RBAC) protect user data and transactions.

System Architecture

The online food ordering platform consists of the following components:

- **Clients:** Web browsers and mobile applications used by customers and restaurant owners.
- **RESTful Web APIs:** Handle communication between frontend applications and backend cloud services.
- **Cloud Application Services:** Manage user registration, restaurant listings, orders, payments, and delivery tracking.
- **Cloud Storage:** Stores menu images, invoices, customer details, and order history securely.
- **Virtual Machines (VMs):** Run backend applications and support cloud services.
- **Managed Kubernetes:** Automatically manages and scales application containers based on user demand.
- **Security Services:** Protect the system using authentication tokens, HTTPS encryption, and RBAC.

Architecture Flow



Role of Cloud Components

Clients

Customers and restaurant staff use:

- Web browsers
- Mobile applications

They can:

- Browse restaurant menus
- Place food orders
- Make online payments
- Track delivery status
- View order history

RESTful Web APIs

RESTful APIs connect the frontend with backend cloud services.

They help to:

- Process food orders.
- Retrieve restaurant menus.
- Update order status.
- Handle payment requests.
- Synchronize data between multiple devices.

Cloud Storage

Cloud storage securely stores:

- Menu images
- Customer information
- Order history
- Payment invoices
- Restaurant details

It provides automatic backup, high availability, and easy scalability.

Virtual Machines (VMs)

Virtual Machines provide the computing environment required to run backend services.

They:

- Host application servers.
- Manage databases and business logic.
- Support stable and isolated execution environments.

- Provide reliable infrastructure for cloud applications.

Managed Kubernetes

Managed Kubernetes simplifies container management by:

- Automatically deploying application containers.
- Scaling services during high demand.
- Restarting failed containers automatically.
- Distributing workloads across multiple servers.
- Ensuring high availability and fault tolerance.

This allows the platform to handle thousands of simultaneous food orders without interruption.

Security Features

The platform protects customer and business data using multiple security mechanisms.

- **Authentication Tokens (JWT/OAuth):** Verify users before granting access.
- **HTTPS:** Encrypts all communication between clients and cloud servers.
- **Role-Based Access Control (RBAC):** Provides different permissions for customers, restaurant owners, delivery agents, and administrators.
- **Encryption:** Protects customer information and payment details stored in the cloud.
- **API Security:** Prevents unauthorized access to backend services.

These measures ensure secure transactions and protect sensitive customer information.

Benefits of the Solution

- Customers can order food anytime using web or mobile applications.
- RESTful APIs provide fast and reliable communication.
- Cloud storage securely stores images, invoices, and customer data.
- Kubernetes automatically scales the application during peak meal hours.
- Virtual Machines provide a reliable infrastructure for backend services.
- High availability ensures continuous service even during heavy traffic.
- Strong security protects customer accounts and payment information.
- The platform can easily expand as the number of restaurants and customers grows.